

PRACTICE PROBLEMS · SOLUTIONS

Linear algebra, solved

Step-by-step solutions to every practice problem in the course notes, numbered as in each lecture. Every answer was checked with exact arithmetic.

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LECTURE 1

Vectors, length and the dot product

1. With $\mathbf{u} = (2, -1)$ and $\mathbf{v} = (-1, 3)$, find $3\mathbf{u} - \mathbf{v}$.

Solution

1. $3\mathbf{u} = (6, -3)$.
2. $3\mathbf{u} - \mathbf{v} = (6 - (-1), -3 - 3) = (7, -6)$.

2. Compute $(1, 0, 2, -1) + 2(0, 3, 1, 1)$ in $\mathbb{R}^4$.

Solution

1. $2(0, 3, 1, 1) = (0, 6, 2, 2)$.
2. Add entry by entry: $(1 + 0, 0 + 6, 2 + 2, -1 + 2) = (1, 6, 4, 1)$.

3. Is $(1, 2, 3)$ a linear combination of $(1, 0, 1)$ and $(0, 1, 1)$?

Solution

1. Write $a(1, 0, 1) + b(0, 1, 1) = (a, b, a + b)$ and match with $(1, 2, 3)$.
2. The first two entries give $a = 1, b = 2$.
3. The third entry needs $a + b = 3$ ✓. Consistent: yes, $(1, 2, 3) = (1, 0, 1) + 2(0, 1, 1)$.

4. Is $(1, 2, 4)$ a linear combination of $(1, 0, 1)$ and $(0, 1, 1)$?

Solution

1. As before, $a(1, 0, 1) + b(0, 1, 1) = (a, b, a + b)$, so $a = 1$ and $b = 2$.
2. The third entry would be $a + b = 3$, but we need 4.
3. Inconsistent: **no**. The point lies off the plane the two vectors span.

5. Find the unit vector in the direction of $(3, -4)$.

Solution

1. Length: $\sqrt{3^2 + (-4)^2} = \sqrt{25} = 5$.
2. Divide by the length: $(\frac{3}{5}, -\frac{4}{5})$. Check: $\frac{9}{25} + \frac{16}{25} = 1$ ✓.

6. Compute $\|(2, -1, 0, 2)\|$.

Solution

1. Square the entries: 4, 1, 0, 4. Add: 9.
2. Take the square root: $\sqrt{9} = 3$.

7. Find a vector of length 10 pointing the same way as $(1, 2, 2)$.

Solution

1. $\|(1, 2, 2)\| = \sqrt{1 + 4 + 4} = 3$, so the unit vector is $(\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$.
2. Multiply by 10: $(\frac{10}{3}, \frac{20}{3}, \frac{20}{3})$.

8. Find k so that $(2, k)$ is perpendicular to $(3, -6)$.

Solution

1. Perpendicular means the dot product is 0: $(2, k) \cdot (3, -6) = 6 - 6k$.
2. $6 - 6k = 0 \Rightarrow k = 1$. Check: $(2, 1) \cdot (3, -6) = 6 - 6 = 0 \checkmark$.

9. Find the angle between $(1, 2, 2)$ and $(2, 0, 0)$.

Solution

1. Dot product: $(1, 2, 2) \cdot (2, 0, 0) = 2$. Lengths: 3 and 2.
2. $\cos \theta = \frac{2}{3 \cdot 2} = \frac{1}{3}$, so $\theta = \cos^{-1} \frac{1}{3} \approx 70.5^\circ$.

10. Let $\mathbf{u} = (2, -1, 4)$ and $\mathbf{v} = (1, 3, -2)$. Find $\mathbf{u} + 2\mathbf{v}$ and $\|\mathbf{u} - \mathbf{v}\|$.

Solution

1. $2\mathbf{v} = (2, 6, -4)$, so $\mathbf{u} + 2\mathbf{v} = (2 + 2, -1 + 6, 4 - 4) = (4, 5, 0)$.
2. $\mathbf{u} - \mathbf{v} = (1, -4, 6)$, so $\|\mathbf{u} - \mathbf{v}\| = \sqrt{1 + 16 + 36} = \sqrt{53}$.

11. Find the unit vector pointing in the *opposite* direction to $(6, -2, 3)$.

Solution

1. Length: $\|(6, -2, 3)\| = \sqrt{36 + 4 + 9} = 7$.
2. Unit vector in the same direction: $(\frac{6}{7}, -\frac{2}{7}, \frac{3}{7})$. The opposite direction is its negative: $(-\frac{6}{7}, \frac{2}{7}, -\frac{3}{7})$.

12. Write $(1, 7)$ as a linear combination of $(1, 2)$ and $(-1, 1)$.

Solution

1. Match entries of $a(1, 2) + b(-1, 1) = (1, 7)$: $a - b = 1$ and $2a + b = 7$.
2. Add the equations: $3a = 8 \Rightarrow a = \frac{8}{3}$. Then $b = a - 1 = \frac{5}{3}$.
3. Check: $\frac{8}{3}(1, 2) + \frac{5}{3}(-1, 1) = (\frac{8-5}{3}, \frac{16+5}{3}) = (1, 7) \checkmark$.

13. Find the angle between $(1, 0, 1)$ and $(0, 1, 1)$.

Solution

1. $(1, 0, 1) \cdot (0, 1, 1) = 0 + 0 + 1 = 1$; both lengths are $\sqrt{2}$.
2. $\cos \theta = \frac{1}{\sqrt{2} \cdot \sqrt{2}} = \frac{1}{2}$, so $\theta = 60^\circ$.

14. Find every vector (x, y) of length 5 that is orthogonal to $(3, 4)$.

Solution

1. Orthogonal to $(3, 4)$: $3x + 4y = 0$, so $(x, y) = t(4, -3)$.
2. Length $\|t(4, -3)\| = 5|t|$. Setting it equal to 5 gives $t = \pm 1$.
3. Answers: $(4, -3)$ and $(-4, 3)$.

LECTURE 2

Projections, cross product, lines and planes

1. Project $(1, 2, 3)$ onto $(1, 1, 1)$, and find the perpendicular part.

Solution

1. $\mathbf{u} \cdot \mathbf{v} = 1 + 2 + 3 = 6$ and $\mathbf{v} \cdot \mathbf{v} = 3$.
2. $\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{6}{3}(1, 1, 1) = (2, 2, 2)$.
3. Perpendicular part: $(1, 2, 3) - (2, 2, 2) = (-1, 0, 1)$. Check: $(-1, 0, 1) \cdot (1, 1, 1) = 0$ ✓.

2. Find the distance from the point $(3, 4)$ to the line through the origin with direction $(2, 1)$.

Solution

1. Project $(3, 4)$ onto $(2, 1)$: $\frac{(3,4) \cdot (2,1)}{(2,1) \cdot (2,1)}(2, 1) = \frac{10}{5}(2, 1) = (4, 2)$.
2. Perpendicular part: $(3, 4) - (4, 2) = (-1, 2)$.
3. The distance is its length: $\sqrt{1 + 4} = \sqrt{5}$.

3. Find the area of the triangle with vertices $(0, 0, 0)$, $(1, 0, 0)$ and $(0, 2, 0)$.

Solution

1. Edge vectors from the origin: $(1, 0, 0)$ and $(0, 2, 0)$.
2. Cross product: $(0 \cdot 0 - 0 \cdot 2, -(1 \cdot 0 - 0 \cdot 0), 1 \cdot 2 - 0 \cdot 0) = (0, 0, 2)$, length 2 = parallelogram area.
3. Triangle area: $\frac{1}{2} \cdot 2 = 1$. Sanity check: a right triangle with legs 1 and 2 has area $\frac{1}{2} \cdot 1 \cdot 2 = 1$ ✓.

4. Find the line through $(1, 0, 2)$ and $(3, 1, 5)$.

Solution

1. Direction: the difference of the points, $(3, 1, 5) - (1, 0, 2) = (2, 1, 3)$.
2. Start at either point: $\mathbf{x} = (1, 0, 2) + t(2, 1, 3)$. (At $t = 1$ you reach $(3, 1, 5)$ ✓.)

5. Find the plane through the origin perpendicular to $(1, -2, 4)$.

Solution

1. The normal $(1, -2, 4)$ gives the coefficients: $x - 2y + 4z = d$.
2. The origin is on the plane, so $d = 0$: $x - 2y + 4z = 0$.

6. Is the point $(3, 2, 2, 5)$ on the line $(1, 0, 2, -1) + t(1, 1, 0, 3)$ in $\mathbb{R}^4$?

Solution

1. Match entry by entry: $1 + t = 3$, $t = 2$, $2 = 2$, $-1 + 3t = 5$.
2. The first, second and fourth all give $t = 2$, and the third holds for any t .
3. One value of t works for every entry, so yes.

7. Find the scalar and vector projections of $\mathbf{u} = (1, 2, 3)$ onto $\mathbf{v} = (2, -1, 2)$.

Solution

1. $\mathbf{u} \cdot \mathbf{v} = 2 - 2 + 6 = 6$ and $\|\mathbf{v}\| = \sqrt{4 + 1 + 4} = 3$.

2. Scalar projection: $\frac{6}{3} = 2$.
3. Vector projection: $\frac{6}{9}(2, -1, 2) = (\frac{4}{3}, -\frac{2}{3}, \frac{4}{3})$.

8. Compute $(1, -1, 2) \times (3, 0, 1)$ and check your answer.

Solution

1. With $\mathbf{u} = (1, -1, 2)$, $\mathbf{v} = (3, 0, 1)$:
 $\mathbf{u} \times \mathbf{v} = ((-1)(1) - 2(0), -[1(1) - 2(3)], 1(0) - (-1)(3)) = (-1, 5, 3)$.
2. Check: $(-1, 5, 3) \cdot (1, -1, 2) = -1 - 5 + 6 = 0$ and $(-1, 5, 3) \cdot (3, 0, 1) = -3 + 0 + 3 = 0 \checkmark$.

9. Find the equation of the plane through $(1, 2, 0)$, $(0, 1, 1)$ and $(2, 0, 1)$.

Solution

1. Edge vectors from $(1, 2, 0)$: $(0, 1, 1) - (1, 2, 0) = (-1, -1, 1)$ and $(2, 0, 1) - (1, 2, 0) = (1, -2, 1)$.
2. Normal:
 $(-1, -1, 1) \times (1, -2, 1) = ((-1)(1) - (1)(-2), -[(-1)(1) - (1)(1)], (-1)(-2) - (-1)(1)) = (1, 2, 3)$.
3. Plane through $(1, 2, 0)$: $1(x - 1) + 2(y - 2) + 3(z - 0) = 0 \Rightarrow x + 2y + 3z = 5$.
4. Check the other points: $0 + 2 + 3 = 5$ and $2 + 0 + 3 = 5 \checkmark$.

10. Find parametric equations for the line where the planes $x + y + z = 6$ and $x - y = 0$ meet.

Solution

1. The line lies in both planes, so its direction is perpendicular to both normals:
 $(1, 1, 1) \times (1, -1, 0) = (0 + 1, -1 - 1, -1 - 1) = (1, 1, -2)$.
2. A common point: set $x = 0$; the second equation gives $y = 0$, then the first gives $z = 6$.
3. Line: $\mathbf{x} = (0, 0, 6) + t(1, 1, -2)$. Check: $t + t + (6 - 2t) = 6$ and $t - t = 0 \checkmark$.

11. Find the distance from $(2, -1, 3)$ to the plane $2x - y + 2z = 1$.

Solution

1. Normal $(2, -1, 2)$, $d = 1$, point $(2, -1, 3)$.
2. Numerator: $|2(2) - (-1) + 2(3) - 1| = |4 + 1 + 6 - 1| = 10$. Denominator: $\sqrt{4 + 1 + 4} = 3$.
3. Distance $\frac{10}{3}$.

LECTURE 3

Systems, row reduction and RREF

1. Solve $x + 2y = 4$, $2x + 4y = 8$.

Solution

$$1. \left[\begin{array}{cc|c} 1 & 2 & 4 \\ 2 & 4 & 8 \end{array} \right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 0 & 0 \end{array} \right].$$

2. Only one pivot; y is free. Let $y = t$: $x = 4 - 2t$. Infinitely many solutions (the two equations are the same line).

2. Solve $x + 2y = 4$, $2x + 4y = 9$.

Solution

$$1. \left[\begin{array}{cc|c} 1 & 2 & 4 \\ 2 & 4 & 9 \end{array} \right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 0 & 1 \end{array} \right].$$

2. The last row says $0 = 1$: no solution (parallel lines).

3. Put $\left[\begin{array}{cc|c} 1 & 3 & 5 \\ 0 & 1 & 2 \end{array} \right]$ into RREF.

Solution

1. It's already in REF; clear the entry above the second pivot:

$$\left[\begin{array}{cc|c} 1 & 3 & 5 \\ 0 & 1 & 2 \end{array} \right] \xrightarrow{R_1 - 3R_2} \left[\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & 1 & 2 \end{array} \right]$$

2. So $x = -1$, $y = 2$.

4. Is $\left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$ in RREF? What's the solution?

Solution

1. Check the rules: pivots (both 1) step to the right, each pivot is alone in its column, and the zero row is at the bottom. **Yes**, it is in RREF.

2. Column 3 has no pivot, so $z = t$ is free.

3. Rows: $x + 2z = 3 \Rightarrow x = 3 - 2t$ and $y = 1$.

5. Solve $2x + y - z = 3$, $x - y + z = 0$, $3x + 2y + z = 8$.

Solution

1. Swap the first two equations so the first pivot is 1, then eliminate:

$$\begin{aligned} & \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 2 & 1 & -1 & 3 \\ 3 & 2 & 1 & 8 \end{array} \right] \xrightarrow{\substack{R_2 - 2R_1 \\ R_3 - 3R_1}} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 3 & -3 & 3 \\ 0 & 5 & -2 & 8 \end{array} \right] \xrightarrow{\frac{1}{3}R_2} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 5 & -2 & 8 \end{array} \right] \\ & \xrightarrow{R_3 - 5R_2} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 3 & 3 \end{array} \right] \end{aligned}$$

2. Back-substitute: $3z = 3 \Rightarrow z = 1$; $y - z = 1 \Rightarrow y = 2$; $x - y + z = 0 \Rightarrow x = 1$.
3. Check in the original first equation: $2 + 2 - 1 = 3 \checkmark$.

6. Find the RREF and the general solution of $x + y + z + w = 4$, $x + 2y + 3z + 4w = 10$.

Solution

1. Row reduce:

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 4 \\ 1 & 2 & 3 & 4 & 10 \end{array} \right] \xrightarrow{R_2 - R_1} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 4 \\ 0 & 1 & 2 & 3 & 6 \end{array} \right] \xrightarrow{R_1 - R_2} \left[\begin{array}{cccc|c} 1 & 0 & -1 & -2 & -2 \\ 0 & 1 & 2 & 3 & 6 \end{array} \right]$$

2. Pivots in the x and y columns; $z = s$ and $w = t$ are free.
3. Rows give $x = -2 + s + 2t$ and $y = 6 - 2s - 3t$.

$$4. \mathbf{x} = \begin{bmatrix} -2 \\ 6 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix}.$$

7. For which a and b does $x + 2y = 3$, $2x + ay = b$ have (i) a unique solution, (ii) no solution, (iii) infinitely many?

Solution

1. $\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & a & b \end{array} \right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & a - 4 & b - 6 \end{array} \right]$.
2. (i) If $a \neq 4$, both columns have pivots: unique solution.
3. (ii) If $a = 4$ and $b \neq 6$, row 2 says $0 = b - 6 \neq 0$: no solution.
4. (iii) If $a = 4$ and $b = 6$, row 2 is all zeros: infinitely many.

8. Solve the homogeneous system $x + y - z = 0$, $2x - y + z = 0$, $x + 2y - 2z = 0$.

Solution

1. Reduce the coefficient matrix:

$$\begin{bmatrix} 1 & 1 & -1 \\ 2 & -1 & 1 \\ 1 & 2 & -2 \end{bmatrix} \xrightarrow{\substack{R_2 - 2R_1 \\ R_3 - R_1}} \begin{bmatrix} 1 & 1 & -1 \\ 0 & -3 & 3 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & -3 & 3 \end{bmatrix}$$

$$\xrightarrow{\substack{R_1 - R_2 \\ R_3 + 3R_2}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

2. $z = t$ is free; the rows give $x = 0$ and $y = z = t$.
3. $\mathbf{x} = t(0, 1, 1)$.

LECTURE 4

Matrix algebra and the inverse

1. Compute $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix}$.

Solution

1. Row view: dot each row with $(5, 6)$.
2. $(1, 2) \cdot (5, 6) = 17$ and $(3, 4) \cdot (5, 6) = 39$, so the answer is $\begin{bmatrix} 17 \\ 39 \end{bmatrix}$.

2. Compute $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$.

Solution

1. Entry (i, j) = row i of the first matrix dotted with column j of the second.
2. $\begin{bmatrix} 1 \cdot 2 + 2 \cdot 1 & 1 \cdot 0 + 2 \cdot 1 \\ 3 \cdot 2 + 4 \cdot 1 & 3 \cdot 0 + 4 \cdot 1 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 10 & 4 \end{bmatrix}$.

3. Write $x - y + 2z = 1$, $3y + z = 0$ in the form $A\mathbf{x} = \mathbf{b}$.

Solution

1. Coefficients go in rows, one column per variable x, y, z ; the missing x in the second equation is a 0.
2. $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

4. Invert $\begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$.

Solution

1. $ad - bc = 4(6) - 7(2) = 10$.
2. Swap a and d , negate b and c : $\begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$. Divide by 10: $\frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$.
3. Check: $\begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$, which divided by 10 is I ✓.

5. Does $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$ have an inverse?

Solution

1. $ad - bc = 2(2) - 4(1) = 0$, so there is **no** inverse.
2. Reason: column 2 is twice column 1, so the matrix squashes the plane onto a line.

6. Let $A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix}$. Compute AB , BA and $A^T B$.

Solution

1. AB : rows of A dotted with columns of B :
 $(1, 2) \cdot (3, 1) = 5$, $(1, 2) \cdot (1, 0) = 1$, $(0, -1) \cdot (3, 1) = -1$, $(0, -1) \cdot (1, 0) = 0$. So $AB = \begin{bmatrix} 5 & 1 \\ -1 & 0 \end{bmatrix}$.

2. BA : $(3, 1) \cdot (1, 0) = 3$, $(3, 1) \cdot (2, -1) = 5$, $(1, 0) \cdot (1, 0) = 1$, $(1, 0) \cdot (2, -1) = 2$. So $BA = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$.

3. $A^T = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$, so $A^T B = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$.

7. Invert $\begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$.

Solution

1. $ad - bc = 2(2) - (-1)(-3) = 1$.

2. Swap a, d and negate b, c : $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$; divide by 1.

3. Check: $\begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \checkmark$.

8. Use $[A \mid I]$ to invert $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.

Solution

1. Row reduce $[A \mid I]$ from the bottom up:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_2 - R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_1 - R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 1 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

2. $A^{-1} = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

9. Solve $XA = B$ for X , where $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$.

Solution

1. Multiply $XA = B$ on the right by A^{-1} : $X = BA^{-1}$.

2. $\det A = 3 - 2 = 1$, so $A^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$.

3. $X = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 6-1 & -4+1 \\ 0-1 & 0+1 \end{bmatrix} = \begin{bmatrix} 5 & -3 \\ -1 & 1 \end{bmatrix}$.

4. Check: $XA = \begin{bmatrix} 5 & -3 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = B \checkmark$.

10. Show that $(A^{-1})^T$ is the inverse of A^T .

Solution

1. We must show $A^T(A^{-1})^T = I$ (and the same in the other order).

2. Use $(XY)^T = Y^T X^T$ backwards: $A^T(A^{-1})^T = (A^{-1}A)^T$.

3. $= I^T = I$. Similarly $(A^{-1})^T A^T = (AA^{-1})^T = I$. So $(A^{-1})^T = (A^T)^{-1}$.

LECTURE 5

Span, independence, basis and dimension

1. Are $(1, 2)$ and $(3, 6)$ independent?

Solution

1. $(3, 6) = 3(1, 2)$: one vector is a multiple of the other, so they're **dependent**.

2. Are $(1, 1, 0)$, $(0, 1, 1)$ and $(1, 2, 1)$ independent?

Solution

1. No two of them are multiples, so check for a combination: $(1, 1, 0) + (0, 1, 1) = (1, 2, 1)$.
 2. So $(1, 1, 0) + (0, 1, 1) - (1, 2, 1) = \mathbf{0}$: **dependent**. (The determinant of the matrix with these columns is 0.)

3. Is $(1, 1, 1)$ in $\text{span}\{(1, 0, 1), (0, 1, 0)\}$?

Solution

1. Solve $a(1, 0, 1) + b(0, 1, 0) = (a, b, a) = (1, 1, 1)$.
 2. This gives $a = 1, b = 1$, and the third entry $a = 1$ ✓. **Yes**.

4. Is $(1, 2, 3)$ in $\text{span}\{(1, 1, 1), (0, 1, 2)\}$?

Solution

1. We need $a(1, 1, 1) + b(0, 1, 2) = (a, a + b, a + 2b) = (1, 2, 3)$.
 2. First entry: $a = 1$. Second: $1 + b = 2 \Rightarrow b = 1$.
 3. Third: $a + 2b = 3$ ✓. Consistent, so **yes**: $(1, 2, 3) = (1, 1, 1) + (0, 1, 2)$.

5. Are $(1, 0, 2)$, $(0, 1, -1)$, $(2, 3, 1)$ independent?

Solution

1. Put the vectors in as columns and expand along row 1: $\begin{vmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 2 & -1 & 1 \end{vmatrix} = 1(1 + 3) - 0 + 2(0 - 2) = 0$.
 2. The determinant is 0, so the vectors are **dependent**.
 3. Find the relation: try $c_1(1, 0, 2) + c_2(0, 1, -1) = (2, 3, 1)$. The first two entries give $c_1 = 2, c_2 = 3$, and the third checks: $4 - 3 = 1$ ✓.

6. Find a basis and the dimension of $\text{span}\{(1, 1, 0), (2, 2, 0), (0, 1, 1), (1, 2, 1)\}$.

Solution

1. Spot the redundant vectors: $(2, 2, 0) = 2(1, 1, 0)$ and $(1, 2, 1) = (1, 1, 0) + (0, 1, 1)$.
 2. Remove them; $(1, 1, 0)$ and $(0, 1, 1)$ are not multiples, so they're independent.
 3. Basis $\{(1, 1, 0), (0, 1, 1)\}$; dimension 2.

7. Find the coordinates of $(2, 3)$ in the basis $\{(1, 1), (1, -1)\}$.

Solution

1. Solve $a(1, 1) + b(1, -1) = (2, 3)$: $a + b = 2$, $a - b = 3$.
2. Add: $2a = 5 \Rightarrow a = \frac{5}{2}$. Subtract: $2b = -1 \Rightarrow b = -\frac{1}{2}$.
3. Check: $\frac{5}{2}(1, 1) - \frac{1}{2}(1, -1) = (2, 3) \checkmark$.

8. For which h is $\{(1, 2, 3), (0, 1, h), (1, 3, 5)\}$ dependent?

Solution

1. Three vectors in $\mathbb{R}^3$ are dependent exactly when the determinant is 0.
2. Expand along row 1: $\begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ 3 & h & 5 \end{vmatrix} = 1(5 - 3h) - 0 + 1(2h - 3) = 2 - h$.
3. Set $2 - h = 0$: $h = 2$.

LECTURE 6

Subspaces, null space, column space and rank

1. Which of these are subspaces of $\mathbb{R}^2$? (a) all (x, y) with $y = 2x$; (b) all (x, y) with $y = 2x + 1$; (c) all (x, y) with $xy \geq 0$.

Solution

- (a) $y = 2x$ is a line through the origin, the span of $(1, 2)$: **subspace**.
- (b) $(0, 0)$ fails $0 = 2(0) + 1$, so the zero vector is missing: **not a subspace**.
- (c) It contains $(0, 0)$, but $(1, 0)$ and $(0, -1)$ are in it while their sum $(1, -1)$ has $xy = -1 < 0$: not closed under addition, **not a subspace**.

2. Find a basis for the plane $x + y + z = 0$, which is the null space of $\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$.

Solution

- One equation, three unknowns: $y = s$ and $z = t$ are free, and $x = -s - t$.
- $$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$
- Basis $\{(-1, 1, 0), (-1, 0, 1)\}$; the plane has dimension 2.

3. A 4×6 matrix has rank 4. What is its nullity?

Solution

- Rank-nullity: **rank** + **nullity** = **number of columns** = 6.
- nullity** = $6 - 4 = 2$.

4. Which are subspaces? (a) $\{(x, y, z) : x = y = z\}$ in $\mathbb{R}^3$; (b) $\{(x, y) : x \geq 0\}$ in $\mathbb{R}^2$; (c) $\{(x, y, z) : z = xy\}$ in $\mathbb{R}^3$.

Solution

- (a) The set is all multiples of $(1, 1, 1)$, i.e. **span** $\{(1, 1, 1)\}$. A span is always a subspace: **yes**.
- (b) $(1, 0)$ is in the set but $(-1)(1, 0) = (-1, 0)$ is not. Not closed under scaling: **no**.
- (c) $(1, 1, 1)$ is in the set ($1 = 1 \cdot 1$), but $2(1, 1, 1) = (2, 2, 2)$ would need $2 = 2 \cdot 2$. **No**.

5. Find a basis for the null space of $\begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ 1 & 3 & 0 \end{bmatrix}$.

Solution

- Row reduce:

$$\begin{aligned} &\begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ 1 & 3 & 0 \end{bmatrix} \xrightarrow{\substack{R_2 - 2R_1 \\ R_3 - R_1}} \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \\ &\xrightarrow{R_1 - 2R_2} \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

2. $x_3 = t$ is free; the rows give $x_1 = 3t$ and $x_2 = -t$.
3. Basis $\{(3, -1, 1)\}$. Check: $A(3, -1, 1) = (3 - 2 - 1, 6 - 4 - 2, 3 - 3 + 0) = \mathbf{0} \checkmark$.

6. Find the rank and nullity of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$, and a basis of its column space.

Solution

1. Row reduce:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow[\substack{R_3 - R_1}]{R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

2. Two pivots (columns 1 and 2): rank 2, nullity $3 - 2 = 1$.
3. Null space: $-y - 2z = 0$ and $x + 2y + 3z = 0$; with $z = 1, y = -2, x = 1$: $\text{span}\{(1, -2, 1)\}$.
4. Column space basis: original columns 1 and 2, $\{(1, 2, 1), (2, 4, 1)\}$.

7. A is 6×4 with rank 4. What is its nullity? How many solutions can $A\mathbf{x} = \mathbf{b}$ have?

Solution

1. Rank-nullity: **nullity** $= 4 - 4 = 0$.
2. So $A\mathbf{x} = \mathbf{0}$ has only $\mathbf{x} = \mathbf{0}$: there is a pivot in every column.
3. If $A\mathbf{x} = \mathbf{b}$ has a solution, it is unique (two solutions would differ by a non-zero null vector). It may have none, because the rank 4 is less than the 6 rows.

LECTURE 7

Midterm review: Units 1–5

1. Do the lines $(1, 0, 0) + t(1, 1, 0)$ and $(0, 1, 1) + s(1, 0, -1)$ intersect?

Solution

1. Set the points equal: $(1 + t, t, 0) = (s, 1, 1 - s)$.
2. The middle entries give $t = 1$; then the first gives $s = 2$.
3. The third entry needs $0 = 1 - s = -1$. Contradiction, so **no intersection**.
4. The directions $(1, 1, 0)$ and $(1, 0, -1)$ aren't parallel, so the lines are **skew**.

2. Solve $x + 2y - z = 1$, $2x + 4y + z = 5$, $x + 2y + 2z = 4$.

Solution

1. Row reduce:

$$\begin{aligned} & \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 4 & 1 & 5 \\ 1 & 2 & 2 & 4 \end{array} \right] \xrightarrow{\substack{R_2 - 2R_1 \\ R_3 - R_1}} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 3 & 3 \end{array} \right] \xrightarrow{\substack{\frac{1}{3}R_2 \\ R_3 - R_2}} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \\ & \xrightarrow{R_1 + R_2} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$

2. $y = t$ is free: $z = 1$ and $x = 2 - 2t$.
3. $\mathbf{x} = (2, 0, 1) + t(-2, 1, 0)$.

3. Is $\{(1, 1, 0, 0), (0, 1, 1, 0), (0, 0, 1, 1), (1, 0, 0, 1)\}$ a basis of $\mathbb{R}^4$?

Solution

1. Look for a relation: $\mathbf{v}_1 - \mathbf{v}_2 = (1, 0, -1, 0)$ and $\mathbf{v}_3 - \mathbf{v}_4 = (-1, 0, 1, 0)$.
2. Adding them gives $\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3 - \mathbf{v}_4 = \mathbf{0}$.
3. A non-trivial relation means the set is dependent, so it is **not a basis**.

4. Find $\begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix}^{-1}$ and use it to solve $5x + 3y = 1$, $3x + 2y = 0$.

Solution

1. $\det = 10 - 9 = 1$, so $A^{-1} = \begin{bmatrix} 2 & -3 \\ -3 & 5 \end{bmatrix}$.
2. $\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} 2 & -3 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$.
3. Check: $5(2) + 3(-3) = 1$ and $3(2) + 2(-3) = 0$ ✓.

LECTURE 8

Linear transformations

1. Is $T(x, y) = (x + y, 2y)$ linear? If so, find its matrix.

Solution

- Each output entry is a fixed combination of x and y , so T is linear.
- $T(\mathbf{e}_1) = T(1, 0) = (1, 0)$ and $T(\mathbf{e}_2) = T(0, 1) = (1, 2)$.
- These are the columns: $\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$.

2. Is $T(x, y) = (x^2, y)$ linear?

Solution

- Test scaling with $\mathbf{e}_1 = (1, 0)$: $T(2\mathbf{e}_1) = T(2, 0) = (4, 0)$.
- But $2T(\mathbf{e}_1) = 2(1, 0) = (2, 0) \neq (4, 0)$. Not linear.

3. Find the matrix that rotates 90° counterclockwise and *then* reflects in the x -axis.

Solution

- Rotation $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, reflection $F = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. "First R , then F " is FR .
- $FR = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$.
- Check: $\mathbf{e}_1 \rightarrow (0, 1) \rightarrow (0, -1)$, the first column $\checkmark$.

4. Find the matrix of the reflection in the line $y = -x$.

Solution

- Reflecting in $y = -x$ sends $(x, y) \mapsto (-y, -x)$.
- $\mathbf{e}_1 = (1, 0) \mapsto (0, -1)$ and $\mathbf{e}_2 = (0, 1) \mapsto (-1, 0)$.
- Columns: $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$.

5. Let $T(x, y, z) = (x - y, y - z, z - x)$. Find its matrix and kernel. Is T onto $\mathbb{R}^3$?

Solution

- Images: $T(\mathbf{e}_1) = (1, 0, -1)$, $T(\mathbf{e}_2) = (-1, 1, 0)$, $T(\mathbf{e}_3) = (0, -1, 1)$, so $A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix}$.
- Kernel: $x - y = 0$, $y - z = 0$ give $x = y = z$: $\ker T = \text{span}\{(1, 1, 1)\}$.
- Nullity 1, so rank $3 - 1 = 2 < 3$: T is **not onto** (and not one-to-one).

6. Write the matrix of rotation by 45° and apply it to $(1, 1)$.

Solution

- Rotation by θ : $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$. With $\cos 45^\circ = \sin 45^\circ = \frac{1}{\sqrt{2}}$: $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$.

2. Apply to $(1, 1)$: $\frac{1}{\sqrt{2}} \begin{bmatrix} 1-1 \\ 1+1 \end{bmatrix} = \begin{bmatrix} 0 \\ \sqrt{2} \end{bmatrix}$.

7. T is linear with $T(1, 1) = (3, 1)$ and $T(1, -1) = (1, 5)$. Find its matrix.

Solution

1. Write $\mathbf{e}_1 = \frac{1}{2}[(1, 1) + (1, -1)]$ and $\mathbf{e}_2 = \frac{1}{2}[(1, 1) - (1, -1)]$.
2. Linearity: $T(\mathbf{e}_1) = \frac{1}{2}[(3, 1) + (1, 5)] = (2, 3)$ and $T(\mathbf{e}_2) = \frac{1}{2}[(3, 1) - (1, 5)] = (1, -2)$.
3. Matrix: $\begin{bmatrix} 2 & 1 \\ 3 & -2 \end{bmatrix}$. Check: $\begin{bmatrix} 2 & 1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \checkmark$.

LECTURE 9

Determinants and Cramer's rule

1. Solve $3x - 2y = 1$, $x + 4y = 5$ by Cramer's rule.

Solution

- $\det A = \begin{vmatrix} 3 & -2 \\ 1 & 4 \end{vmatrix} = 12 + 2 = 14.$
- $x = \frac{1}{14} \begin{vmatrix} 1 & -2 \\ 5 & 4 \end{vmatrix} = \frac{4+10}{14} = 1$ and $y = \frac{1}{14} \begin{vmatrix} 3 & 1 \\ 1 & 5 \end{vmatrix} = \frac{15-1}{14} = 1.$
- Check: $3 - 2 = 1$ and $1 + 4 = 5$ ✓.

2. Can Cramer's rule solve $x + 2y = 4$, $2x + 4y = 8$? Explain.

Solution

- $\det A = \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 4 - 4 = 0$, and Cramer's rule divides by $\det A$, so it can't be used.
- The second equation is twice the first, so the system actually has infinitely many solutions; row reduction finds them.

3. Compute $\begin{vmatrix} 2 & 0 & 1 \\ 1 & 3 & -1 \\ 0 & 5 & 4 \end{vmatrix}.$

Solution

- Expand along row 1 (it has a zero): $\begin{vmatrix} 2 & 0 & 1 \\ 1 & 3 & -1 \\ 0 & 5 & 4 \end{vmatrix} = 2 \begin{vmatrix} 3 & -1 \\ 5 & 4 \end{vmatrix} - 0 + 1 \begin{vmatrix} 1 & 3 \\ 0 & 5 \end{vmatrix}.$
- $= 2(12 + 5) + 1(5 - 0) = 34 + 5 = 39.$

4. Compute $\begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{vmatrix}.$

Solution

- Subtract each row from the one below, starting at the bottom ($R_4 - R_3$, then $R_3 - R_2$, then $R_2 - R_1$). Replacements don't change the determinant.
- The result is $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, which is triangular.
- Product of the diagonal: $1 \cdot 1 \cdot 1 \cdot 1 = 1.$

5. Solve $3x + 2y = 7$, $x - y = -1$ by Cramer's rule.

Solution

- $\det A = \begin{vmatrix} 3 & 2 \\ 1 & -1 \end{vmatrix} = -3 - 2 = -5.$

$$2. \quad x = \frac{1}{-5} \begin{vmatrix} 7 & 2 \\ -1 & -1 \end{vmatrix} = \frac{-7+2}{-5} = 1.$$

$$3. \quad y = \frac{1}{-5} \begin{vmatrix} 3 & 7 \\ 1 & -1 \end{vmatrix} = \frac{-3-7}{-5} = 2.$$

$$4. \quad \text{Check: } 3 + 4 = 7 \text{ and } 1 - 2 = -1 \checkmark.$$

6. A and B are 4×4 with $\det A = 3$ and $\det B = -2$. Find $\det(AB^{-1})$, $\det(-A)$, $\det(A^T B^2)$.

Solution

$$1. \quad \det(B^{-1}) = \frac{1}{-2}, \text{ so } \det(AB^{-1}) = 3 \cdot \left(-\frac{1}{2}\right) = -\frac{3}{2}.$$

$$2. \quad \det(-A) = (-1)^4 \det A = 3 \text{ (each of the 4 rows is multiplied by } -1\text{)}.$$

$$3. \quad \det(A^T B^2) = \det A (\det B)^2 = 3 \cdot 4 = 12.$$

LECTURE 10

Eigenvalues and eigenvectors

1. Find the eigenvalues and eigenvectors of $\begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}$.

Solution

1. Characteristic polynomial: $(4 - \lambda)(1 - \lambda) - (-2)(1) = \lambda^2 - 5\lambda + 6 = (\lambda - 2)(\lambda - 3)$.
2. $\lambda = 2$: $A - 2I = \begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix}$ gives $x = y$: $(1, 1)$.
3. $\lambda = 3$: $A - 3I = \begin{bmatrix} 1 & -2 \\ 1 & -2 \end{bmatrix}$ gives $x = 2y$: $(2, 1)$.
4. Check: $A(1, 1) = (2, 2)$ and $A(2, 1) = (6, 3)$ ✓.

2. Show that $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ has no real eigenvalues. What does it do geometrically?

Solution

1. $\det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 \\ -1 & -\lambda \end{vmatrix} = \lambda^2 + 1$.
2. $\lambda^2 + 1 \geq 1 > 0$ for every real λ , so there are no real eigenvalues.
3. The matrix sends $(1, 0) \rightarrow (0, -1)$ and $(0, 1) \rightarrow (1, 0)$: a 90° clockwise rotation, which turns every direction.

3. Find the eigenvalues and an eigenvector for each of $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$.

Solution

1. Upper triangular: eigenvalues 1, 2, 3.
2. $\lambda = 1$: $A - I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ gives $y = z = 0$: $(1, 0, 0)$.
3. $\lambda = 2$: $A - 2I = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ gives $x = z = 0$: $(0, 1, 0)$.
4. $\lambda = 3$: $A - 3I = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ gives $x = 0, y = z$: $(0, 1, 1)$.

4. Show that A and A^T have the same eigenvalues.

Solution

1. $A^T - \lambda I = (A - \lambda I)^T$ because $I^T = I$.
2. A matrix and its transpose have the same determinant, so $\det(A^T - \lambda I) = \det(A - \lambda I)$.
3. Same characteristic polynomial, so the same eigenvalues. (The eigenvectors can differ.)

LECTURE 11

Diagonalization

1. Diagonalize $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$.

Solution

1. $\det(A - \lambda I) = (1 - \lambda)^2 - 4 = \lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1)$.
2. $\lambda = 3: A - 3I = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$ gives $(1, 1)$. $\lambda = -1: A + I = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$ gives $(1, -1)$.
3. $P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, $D = \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}$. Check: trace $2 = 3 + (-1)$, $\det -3 = 3(-1)$ ✓.

2. Is $\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ diagonalizable?

Solution

1. Upper triangular: $\lambda = 2$ twice.
2. $A - 2I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has rank 1, so the eigenspace is 1-dimensional: only the direction $(1, 0)$.
3. We need 2 independent eigenvectors and have only 1: **not diagonalizable**.

3. Diagonalize $A = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}$ (from Lecture 10) and check $AP = PD$.

Solution

1. From Lecture 10: $\lambda = 2$, $(1, 1)$ and $\lambda = 3$, $(2, 1)$.
2. $P = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$.
3. Check: $AP = [A(1, 1) \ A(2, 1)] = \begin{bmatrix} 2 & 6 \\ 2 & 3 \end{bmatrix}$ and $PD = [2(1, 1) \ 3(2, 1)] = \begin{bmatrix} 2 & 6 \\ 2 & 3 \end{bmatrix}$ ✓.

4. Use the previous answer to compute A^5 .

Solution

1. $\det P = 1 - 2 = -1$, so $P^{-1} = -\begin{bmatrix} 1 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$.
2. $D^5 = \begin{bmatrix} 32 & 0 \\ 0 & 243 \end{bmatrix}$, and $PD^5 = \begin{bmatrix} 32 & 486 \\ 32 & 243 \end{bmatrix}$.
3. $A^5 = PD^5P^{-1} = \begin{bmatrix} -32 + 486 & 64 - 486 \\ -32 + 243 & 64 - 243 \end{bmatrix} = \begin{bmatrix} 454 & -422 \\ 211 & -179 \end{bmatrix}$.

5. Which are diagonalizable: $\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$, $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$?

Solution

1. $\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$: $\lambda = 3$ twice. $A - 3I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has nullity 1, so only one eigenvector direction $(1, 0)$: **not diagonalizable**.

2. $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$ is already diagonal ($P = I$): **diagonalizable**. Every non-zero vector is an eigenvector.

6. A population moves between two regions with $A = \begin{bmatrix} \frac{4}{5} & \frac{3}{10} \\ \frac{1}{5} & \frac{7}{10} \end{bmatrix}$. Find the long-run split.

Solution

1. Steady state: $(A - I)\mathbf{x} = \mathbf{0}$ with $A - I = \begin{bmatrix} -\frac{1}{5} & \frac{3}{10} \\ \frac{1}{5} & -\frac{3}{10} \end{bmatrix}$.
2. $-0.2x + 0.3y = 0 \Rightarrow 2x = 3y$, so $\mathbf{x} = (3, 2)$.
3. Scale to total 1: $(\frac{3}{5}, \frac{2}{5})$: 60% and 40%. (The other eigenvalue, $1.5 - 1 = 0.5$, dies out.)

LECTURE 12

Orthonormal bases, Gram–Schmidt and projections

1. Apply Gram–Schmidt to $(3, 4)$, $(1, 0)$, then make the result orthonormal.

Solution

- $\mathbf{v}_1 = (3, 4)$, $\|\mathbf{v}_1\|^2 = 25$.
- $\mathbf{x}_2 \cdot \mathbf{v}_1 = 3$, so $\mathbf{v}_2 = (1, 0) - \frac{3}{25}(3, 4) = (\frac{16}{25}, -\frac{12}{25})$, which points along $(4, -3)$. Check: $12 - 12 = 0$ ✓.
- Both $(3, 4)$ and $(4, -3)$ have length 5: orthonormal basis $(\frac{3}{5}, \frac{4}{5})$, $(\frac{4}{5}, -\frac{3}{5})$.

2. Is $Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ orthogonal? What is Q^{-1} ?

Solution

- Columns $\frac{1}{\sqrt{2}}(1, 1)$ and $\frac{1}{\sqrt{2}}(1, -1)$: each has length $\sqrt{\frac{1}{2} + \frac{1}{2}} = 1$, and their dot product is $\frac{1}{2} - \frac{1}{2} = 0$.
- Orthonormal columns, so Q is orthogonal and $Q^{-1} = Q^T$.
- Q is symmetric, so $Q^T = Q$: Q is its own inverse.

3. Show that $\{(1, 2, -1), (1, 0, 1), (1, -1, -1)\}$ is orthogonal and write $(2, 1, 0)$ in this basis.

Solution

- Dot products:
 $(1, 2, -1) \cdot (1, 0, 1) = 0$, $(1, 2, -1) \cdot (1, -1, -1) = 1 - 2 + 1 = 0$, $(1, 0, 1) \cdot (1, -1, -1) = 0$.
 Orthogonal.
- Squared lengths: 6, 2, 3. Dot products with $\mathbf{x} = (2, 1, 0)$: 4, 2, 1.
- $c_1 = \frac{4}{6} = \frac{2}{3}$, $c_2 = \frac{2}{2} = 1$, $c_3 = \frac{1}{3}$.
- Check: $\frac{2}{3}(1, 2, -1) + (1, 0, 1) + \frac{1}{3}(1, -1, -1) = (2, 1, 0)$ ✓.

4. Apply Gram–Schmidt to $(1, 1, 0)$, $(0, 1, 1)$, $(1, 0, 1)$.

Solution

- $\mathbf{v}_1 = (1, 1, 0)$, $\|\mathbf{v}_1\|^2 = 2$.
- $\mathbf{x}_2 \cdot \mathbf{v}_1 = 1$: $\mathbf{v}_2 = (0, 1, 1) - \frac{1}{2}(1, 1, 0) = (-\frac{1}{2}, \frac{1}{2}, 1)$; scale by 2: $(-1, 1, 2)$, $\|\mathbf{v}_2\|^2 = 6$.
- $\mathbf{x}_3 \cdot \mathbf{v}_1 = 1$ and $\mathbf{x}_3 \cdot \mathbf{v}_2 = -1 + 0 + 2 = 1$: $\mathbf{v}_3 = (1, 0, 1) - \frac{1}{2}(1, 1, 0) - \frac{1}{6}(-1, 1, 2) = (\frac{2}{3}, -\frac{2}{3}, \frac{2}{3})$; scale: $(1, -1, 1)$.
- Check: all three pairwise dot products are 0 ✓.

5. Project $(1, 1, 1)$ onto $U = \text{span}\{(1, 2, 0), (-2, 1, 0)\}$ and find the distance to U .

Solution

- The spanning vectors are orthogonal: $(1, 2, 0) \cdot (-2, 1, 0) = 0$.
- Coefficients: $\frac{(1,1,1) \cdot (1,2,0)}{5} = \frac{3}{5}$ and $\frac{(1,1,1) \cdot (-2,1,0)}{5} = -\frac{1}{5}$.
- $\text{proj}_U(1, 1, 1) = \frac{3}{5}(1, 2, 0) - \frac{1}{5}(-2, 1, 0) = (1, 1, 0)$.
- Leftover $(0, 0, 1)$, so the distance is 1.

6. Find an orthonormal basis of the plane $x + y + z = 0$.

Solution

1. The plane is **null** $\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$, which is 2-dimensional. One vector in it: $(1, -1, 0)$.
2. Gram-Schmidt with a second vector $(1, 0, -1)$: $(1, 0, -1) - \frac{1}{2}(1, -1, 0) = (\frac{1}{2}, \frac{1}{2}, -1)$; scale to $(1, 1, -2)$. It satisfies $1 + 1 - 2 = 0$ and is orthogonal to $(1, -1, 0)$.
3. Normalize: $\frac{1}{\sqrt{2}}(1, -1, 0)$ and $\frac{1}{\sqrt{6}}(1, 1, -2)$.

LECTURE 13

Symmetric matrices and quadratic forms

1. Orthogonally diagonalize $\begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$.

Solution

- Trace 7, det $10 - 4 = 6$: $\lambda^2 - 7\lambda + 6 = (\lambda - 6)(\lambda - 1)$.
- $\lambda = 6$: $A - 6I = \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix}$ gives $(2, 1)$. $\lambda = 1$: $A - I = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$ gives $(1, -2)$.
- They're orthogonal ($2 - 2 = 0$); normalize by $\sqrt{5}$: $Q = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$, $D = \begin{bmatrix} 6 & 0 \\ 0 & 1 \end{bmatrix}$.

2. Orthogonally diagonalize $\begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix}$.

Solution

- Characteristic polynomial: $\lambda^2 - 9\lambda + (18 - 4) = (\lambda - 7)(\lambda - 2)$.
- $\lambda = 7$: $A - 7I = \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix}$ gives $(2, 1)$. $\lambda = 2$: $A - 2I = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$ gives $(1, -2)$.
- Orthogonal ($2 - 2 = 0$); normalize by $\sqrt{5}$: $Q = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$, $D = \begin{bmatrix} 7 & 0 \\ 0 & 2 \end{bmatrix}$.

3. Orthogonally diagonalize $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$.

Solution

- Each row of the matrix sums to 2, so $(1, 1, 1)$ is an eigenvector with $\lambda = 2$.
- The trace is 0, so the other two eigenvalues sum to -2 . $A + I$ is the all-ones matrix, which has rank 1, so $\lambda = -1$ has a 2-dimensional eigenspace: the plane $x + y + z = 0$.
- Choose an orthogonal basis of that plane (as in Lecture 12): $(1, -1, 0)$ and $(1, 1, -2)$.

- Normalize and use as columns: $Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \end{bmatrix}$, $D = \text{diag}(2, -1, -1)$.

4. Classify $q = 2x^2 + 6xy + 2y^2$.

Solution

- $A = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$ (half of 6 goes in each off-diagonal entry).
- Characteristic polynomial: $(2 - \lambda)^2 - 9 = 0 \Rightarrow 2 - \lambda = \pm 3 \Rightarrow \lambda = 5, -1$.
- One positive and one negative eigenvalue: **indefinite**.

5. Identify the curve $3x^2 + 2xy + 3y^2 = 8$.

Solution

1. $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$: eigenvalues 3 ± 1 , i.e. 4 (along $(1, 1)$) and 2 (along $(1, -1)$).
2. In those coordinates: $4u^2 + 2v^2 = 8$.
3. Divide by 8: $\frac{u^2}{2} + \frac{v^2}{4} = 1$. An ellipse with semi-axes $\sqrt{2}$ and 2.

LECTURE 14

Least squares (with MATLAB)

1. Fit $y = a + bx$ to $(0, 0), (1, 1), (2, 1)$.

Solution

$$1. A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

$$2. A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, A^T \mathbf{y} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

$$3. \text{Solve } 3a + 3b = 2, 3a + 5b = 3: \text{ subtract to get } 2b = 1 \Rightarrow b = \frac{1}{2}; \text{ then } 3a = 2 - \frac{3}{2} \Rightarrow a = \frac{1}{6}.$$

$$4. y = \frac{1}{6} + \frac{1}{2}x. \text{ Check: errors } (-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}) \text{ sum to } 0 \checkmark.$$

2. Fit $y = a + bx$ to $(0, 1), (1, 1), (2, 2), (3, 2)$.

Solution

$$1. A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \end{bmatrix}.$$

$$2. A^T A = \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}, A^T \mathbf{y} = \begin{bmatrix} 6 \\ 11 \end{bmatrix}.$$

$$3. \text{Solve } 4a + 6b = 6, 6a + 14b = 11: \text{ multiply the first by } 1.5 \text{ to get } 6a + 9b = 9, \text{ subtract: } 5b = 2 \Rightarrow b = \frac{2}{5}; \text{ then } 4a = 6 - \frac{12}{5} \Rightarrow a = \frac{9}{10}.$$

$$4. y = 0.9 + 0.4x.$$

3. Find the least-squares solution of $x = 1, y = 2, x + y = 4$.

Solution

$$1. A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

$$2. A^T A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, A^T \mathbf{b} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

$$3. \text{Solve } 2x + y = 5, x + 2y = 6: \text{ from the first } y = 5 - 2x, \text{ so } x + 10 - 4x = 6 \Rightarrow x = \frac{4}{3} \text{ and } y = \frac{7}{3}.$$

4. Fit a line through the origin, $y = bx$, to $(1, 2), (2, 3), (3, 7)$.

Solution

$$1. A \text{ is the single column of } x\text{-values } (1, 2, 3), \text{ so the normal equation is } (A^T A) b = A^T \mathbf{y} \text{ with numbers:}$$

$$2. A^T A = 1 + 4 + 9 = 14 \text{ and } A^T \mathbf{y} = 1(2) + 2(3) + 3(7) = 29.$$

$$3. b = \frac{29}{14}.$$

5. Write MATLAB commands to fit $y = a + bx + cx^2$ to data vectors `x` and `y` (columns).

Solution

1. Each row of A is $(1, x_i, x_i^2)$.
2. `A = [ones(size(x)) x x.^2];` builds it (`.^` squares each entry).
3. `coef = A\y` returns a, b, c in order.

LECTURE 15

Final review I: Units 6–8

1. Find the matrix that rotates 90° counterclockwise and then doubles every length. What are its determinant and real eigenvalues?

Solution

1. Rotation by 90° : $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Doubling every length: multiply by 2.
2. The combined matrix: $\begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$.
3. $\det = 0 + 4 = 4$, so areas scale by 4.
4. $\det(A - \lambda I) = \lambda^2 + 4 > 0$: no real eigenvalues (every direction rotates).

2. Is $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix}$ invertible?

Solution

1. Expand along row 1: $\begin{vmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 2 & 0 & 1 \end{vmatrix} = 1(1 - 0) - 2(0 - 4) + 0 = 1 + 8 = 9$.
2. Non-zero determinant, so invertible.

3. Diagonalize $\begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$.

Solution

1. Upper triangular: $\lambda = 2, -1$.
2. $\lambda = 2$: $A - 2I = \begin{bmatrix} 0 & 3 \\ 0 & -3 \end{bmatrix}$ gives $y = 0$: $(1, 0)$.
3. $\lambda = -1$: $A + I = \begin{bmatrix} 3 & 3 \\ 0 & 0 \end{bmatrix}$ gives $x = -y$: $(1, -1)$.
4. $P = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$.

4. A matrix has eigenvector $(1, 1)$ with eigenvalue 3 and $(1, -1)$ with eigenvalue 1. Find it.

Solution

1. $P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, $D = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$, and $P^{-1} = \frac{1}{-2} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$.
2. $PD = \begin{bmatrix} 3 & 1 \\ 3 & -1 \end{bmatrix}$.
3. $A = PDP^{-1} = \frac{1}{2} \begin{bmatrix} 3 & 1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

LECTURE 16

Final review II: Units 9–11 and the whole course

1. Find an orthonormal basis for the column space of $\begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Solution

- Columns $\mathbf{x}_1 = (1, 1, 0)$, $\mathbf{x}_2 = (1, 0, 1)$. $\mathbf{v}_1 = \mathbf{x}_1$, $\|\mathbf{v}_1\|^2 = 2$.
- $\mathbf{x}_2 \cdot \mathbf{v}_1 = 1$, so $\mathbf{v}_2 = (1, 0, 1) - \frac{1}{2}(1, 1, 0) = (\frac{1}{2}, -\frac{1}{2}, 1)$; scale to $(1, -1, 2)$.
- Normalize: $\frac{1}{\sqrt{2}}(1, 1, 0)$ and $\frac{1}{\sqrt{6}}(1, -1, 2)$.

2. Orthogonally diagonalize $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.

Solution

- Characteristic polynomial: $(1 - \lambda)^2 - 1 = \lambda(\lambda - 2)$, so $\lambda = 2, 0$.
- $\lambda = 2$: $A - 2I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$ gives $(1, 1)$. $\lambda = 0$: $A\mathbf{x} = \mathbf{0}$ gives $(1, -1)$.
- $Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$.

3. Fit $\mathbf{y} = \mathbf{a} + \mathbf{b}\mathbf{x}$ to $(-1, 0)$, $(0, 1)$, $(1, 3)$.

Solution

- $A = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$.
- $A^T A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$, $A^T \mathbf{y} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$.
- $3a = 4 \Rightarrow a = \frac{4}{3}$ and $2b = 3 \Rightarrow b = \frac{3}{2}$: $y = \frac{4}{3} + \frac{3}{2}x$.

4. True or false: (a) an orthogonal matrix is always symmetric; (b) the projection of $\mathbf{b}$ onto $\text{col } A$ is $A\hat{\mathbf{x}}$; (c) Gram-Schmidt applied to a dependent set produces a zero vector.

Solution

- (a) A rotation $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ is orthogonal but not symmetric. **False**.
- (b) $\hat{\mathbf{x}}$ is chosen so that $A\hat{\mathbf{x}}$ is the point of $\text{col } A$ closest to $\mathbf{b}$, which is the projection. **True**.
- (c) If $\mathbf{x}_k$ is a combination of earlier vectors, subtracting its projections removes everything: $\mathbf{v}_k = \mathbf{0}$. **True**.